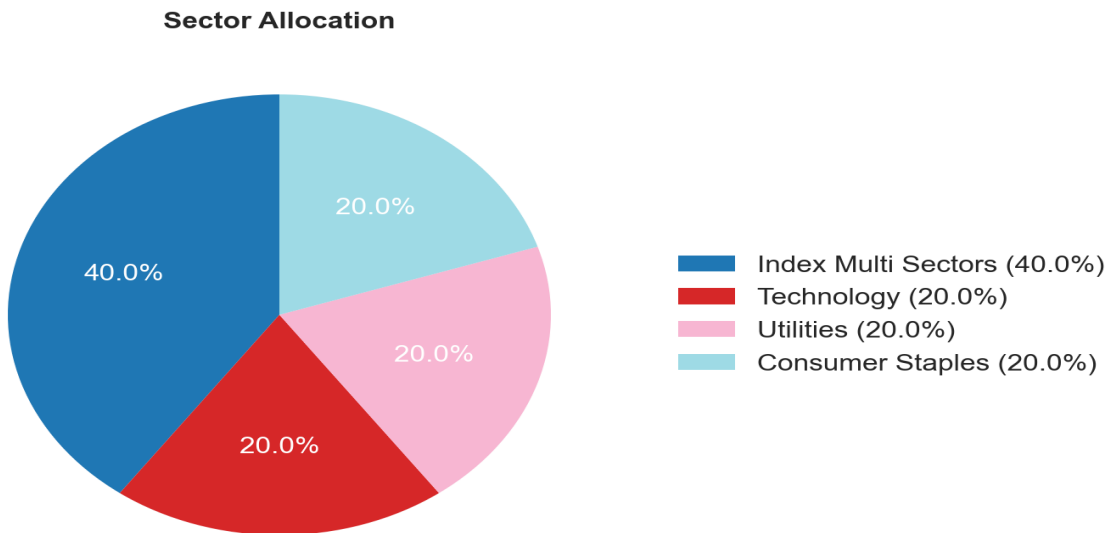


Portfolio rotation report

Date of extraction	2025-12-08 01:24
Data window	2020-12-31 → 2021-12-31
Frequency	Daily — 253 observations. Sufficient depth: short- and long-term indicators are statistically reliable. Warning: the observation window exceeds the typical interpretability range for this frequency (+26.5% above the recommended upper bound). Results may blend multiple market regimes, reducing economic signal clarity.
Tickers	QQQ, SPY, XLK, XLP, XLU
Indicators	Short term: 25 days volatility, 50 days return, 75 days momentum Long term: 62 days volatility, 100 days return, 225 days momentum

1. Portfolio Composition

- XLK (Technology Select Sector SPDR Fund — Technology)
- XLU (Utilities Select Sector SPDR Fund — Utilities)
- XLP (Consumer Staples Select Sector SPDR Fund — Consumer Staples)
- SPY (SPDR S&P; 500 ETF — Index Multi Sectors)
- QQQ (Invesco QQQ Trust — Index Multi Sectors)



Macro Profile Insight

Macro coverage

- Partial coverage: several macro themes remain unrepresented.
- Broad balance across market capitalizations: diversification achieved across the size spectrum.

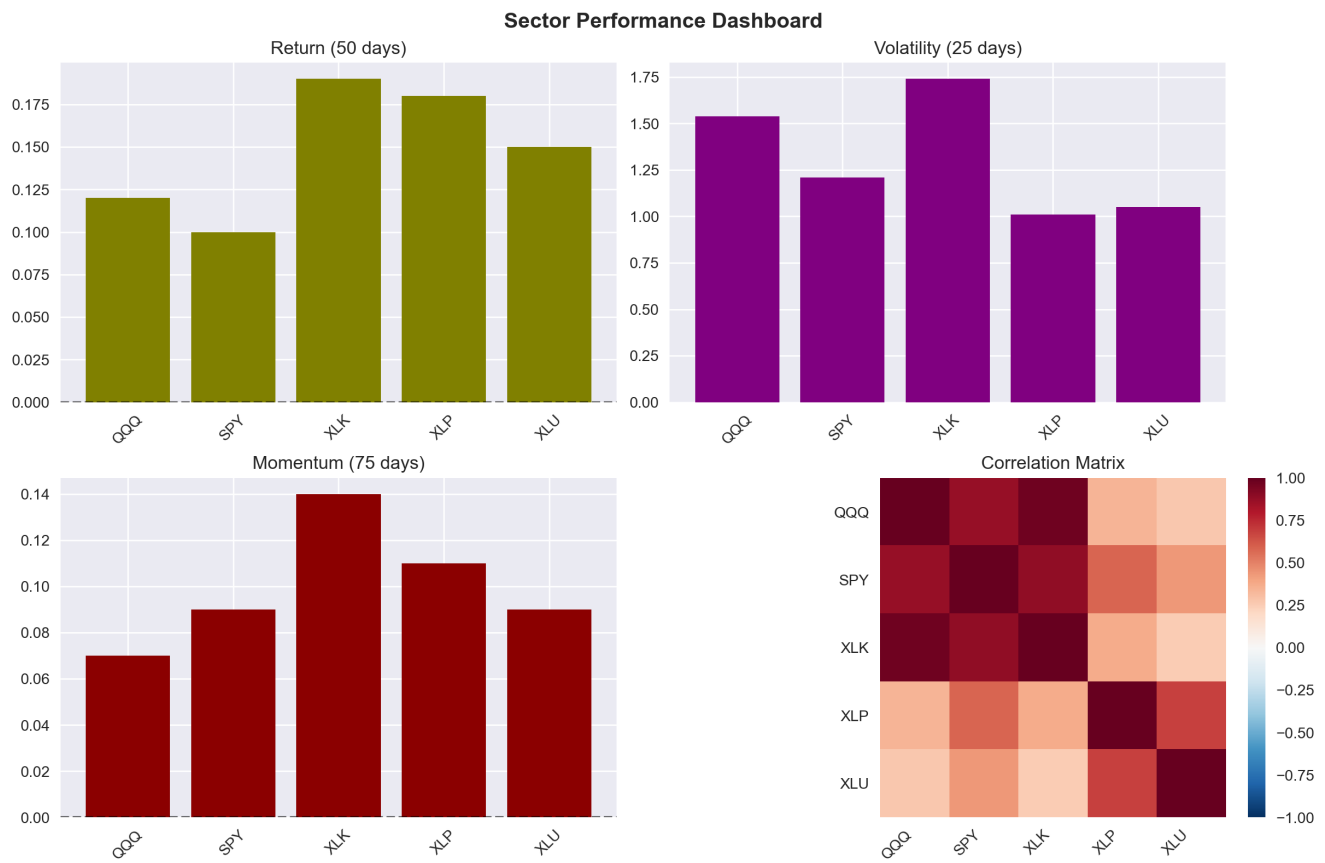
Structural properties

- Concentration risk: portfolio heavily exposed to etf instruments.
- Liquidity stance: pockets of thin/illiquid names; robust median dollar volume.
- Mixed beta structure: diversified sensitivity across holdings.
- Balanced volatility structure: risk evenly distributed across holdings.

Additional comments

- Significant sector gaps: broad portions of the economy remain uncovered.

2. Sector Performance Visualization



Top Positive Correlations

100.0% of portfolio

Pair	Correlation
QQQ / XLK	0.97
SPY / XLK	0.89
QQQ / SPY	0.87
XLP / XLU	0.68
SPY / XLP	0.58

Overall, average cross-sector correlation was 0.57, ranging from 0.25 to 0.97. This suggests elevated co-movement — diversification benefits are partially reduced. Correlation dispersion remains meaningful, indicating both tightly and weakly linked sectors. Strongest co-movement between QQQ and XLK (0.97); weakest between XLK and XLU (0.25). Mixed regime: both market-wide and sector-specific forces are at play.

3. Key Performance Indicators

Ticker	R (%)	V (%)	RAR	M (%)	Signal	Regime	Liquidity	Commentary
QQQ	0.12	1.54	0.079	0.07	neutral	subdued	liquid	mixed short-term dynamics, signal unclear or in transition, low volatility — calm market tone, with ample liquidity.
SPY	0.10	1.21	0.086	0.09	neutral	subdued	liquid	mixed short-term dynamics, signal unclear or in transition, low volatility — calm market tone, with ample liquidity.
XLK	0.19	1.74	0.11	0.14	neutral	normal	thin	mixed short-term dynamics, signal unclear or in transition, normal volatility regime, under limited liquidity.
XLP	0.18	1.01	0.181	0.11	neutral	normal	thin	mixed short-term dynamics, signal unclear or in transition, normal volatility regime, under limited liquidity.
XLU	0.15	1.05	0.143	0.09	neutral	normal	thin	mixed short-term dynamics, signal unclear or in transition, normal volatility regime, under limited liquidity.

Top by Average Return

Ticker	AvgReturn(%)
XLK	0.19
XLP	0.18
XLU	0.15
QQQ	0.12
SPY	0.1

Top by Volatility

Ticker	Volatility(%)
XLK	1.74
QQQ	1.54
SPY	1.21
XLU	1.05
XLP	1.01

Top by Momentum

Ticker	Momentum(%)
XLK	0.14
XLP	0.11
SPY	0.09
XLU	0.09
QQQ	0.07

3.b Rotation & Stress Metrics

Ticker	ΔReturn (pp)	Momentum Ratio	Signal	Volatility Ratio	Beta	ADV10 USD	Interpretation
QQQ	0.02	0.70	0.041	1.36	1.18	53.05 M	stable performance, momentum fading, volatility expanding, within defensive profile, under thin liquidity regime.
SPY	0.00	0.87	0.048	1.42	1.00	72.91 M	stable performance, momentum stable, volatility expanding, within defensive profile, under thin liquidity regime.
XLK	0.07	1.11	0.066	1.40	1.33	16.33 M	stable performance, momentum stable, volatility expanding, within defensive profile, under thin liquidity regime.
XLP	0.08	1.10	0.052	1.37	0.25	16.28 M	stable performance, momentum stable, volatility expanding, within defensive profile, under thin liquidity regime.
XLU	0.05	0.90	0.047	1.16	0.42	19.35 M	stable performance, momentum stable, volatility normalizing, within defensive profile, under thin liquidity regime.

■ Light-impact observations

For **QQQ, SPY**: *Volatility compression* — subdued volatility; a breakout setup may be forming. (avg ann. return 0.1%, momentum 0.1%, vol 1.4%)

For **XLK, XLP, XLU**: *Range / noise* — no clear direction; rotating momentum and mid-range volatility. (avg ann. return 0.2%, momentum 0.1%, vol 1.3%)